

42088 – Industrial Project

Project Status Report

Milestone 4 – Construction II

Project name:	Automatic, universal push-button actuator, equipped with force gauge
Customer:	Altice Labs
Customer supervisor:	Eng. Hélder Alves Eng. Nuno Balseiro
Date issued:	18/12/2024
Team members:	<u>Coordinator:</u> Cláudia Almeida, claudiasalmeida@ua.pt , 914102617. <u>Other team members:</u> Rafael Alves, rafael01@ua.pt , 937353627. Gonçalo Peralta, goncaloperalta@ua.pt , 916828627. Pedro Matias, pedrodsmatias@ua.pt , 917126548.
Supervisor:	Professor Pedro Fonseca
Class supervisor:	Professor Pedro Fonseca

Management summary:

The project is on track to meet its delivery deadlines. The team has successfully completed the initial construction phase and is now transitioning into its final phase of the project where a minimum viable product is expected to be achieved. At this stage, the PCB board has been soldered and tested, showing correct operation. Both the linear actuator and the force sensor were tested individually. The actuator behaved correctly to the commands outlined in Python code by the Raspberry Pi, adjusting its excursion and speed of movement according to the current limitation imposed and the duty cycle programmed. The force sensor correctly measured manually applied forces and provided feedback. The mechanical claw has been printed and has matched the expected result. It is currently undergoing some manual changes and screw positioning to provide more stability for the components it will support. The most recent update includes an extra added support for the force sensor to add stability and be centred with the actuator, to correctly measure the force exerted.

As we move forward our focus will be on improving the overall coupling of all physical parts to a more elegant and discreet version and improve the web interface and data processing of the API in relation to the Raspberry Pi.

Architecture

The architecture of the system consists of three main components: the **Control Unit**, the **Control Electric Circuit**, and the **Mechanical Actuation**.

- The **Control unit** is managed by a Raspberry Pi that runs the control software and serves as the central processing component for the system. The Raspberry Pi generates a PWM (Pulse Width Modulation) signal to regulate the actuator's movement, allowing for controlled button pressing with variable force. It also maintains a Secure Shell (SSH) connection with the DUT, allowing it to detect feedback from the device in real time, such as system responses or status changes upon a button click. For force measurement, the Raspberry Pi reads data from the force sensor using I2C/SPI protocols, ensuring accurate and timely data transfer from the sensor to the control software.
- The **Control electric circuit** amplifies the PWM signal generated by the Raspberry Pi, ensuring sufficient power is supplied to the actuator. This circuit includes an **H-Bridge**, allowing current to flow in both directions as required to control the actuator's movement accurately. The circuit, powered by a 12V, 3A DC power source, will provide the energy needed to drive the actuator with stability and precision, allowing for both pressing and releasing movements, required for testing various types of buttons.

Figure 1 presents PCB, assembled with the support of an Altice Labs engineer. A total of five PCBs were manufactured; one was fully soldered by the engineer to ensure high-quality assembly, while another was soldered by the team as a learning exercise to develop soldering skills. For testing and development, the PCB soldered by the engineer was utilized to minimize the risk of issues arising from improperly soldered or damaged components.

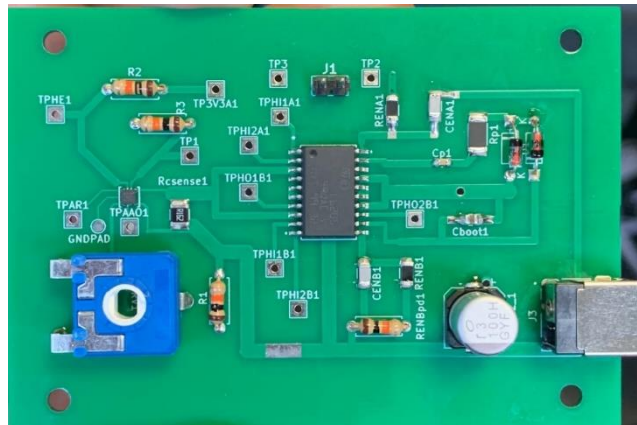


Figure 1. Soldered printed circuit board.

- The **Mechanical Actuation** assembly is designed to interface directly with the DUT, positioning the actuator over the buttons to be tested. This component includes a clamp that securely attaches to the DUT, providing flexibility and adaptability for different button placements and device sizes. Mounted within the clamp is a linear actuator that performs the physical pressing of the buttons, with a force sensor placed behind the actuator to capture the exact force applied during each press. This assembly connects to a

PCB board, which relays signals from the force sensor back to the Control Unit, ensuring accurate control and data capture throughout the testing process.

Figure 2 and Figure 3 show the mechanical part isolated and applied to the test device, respectively.

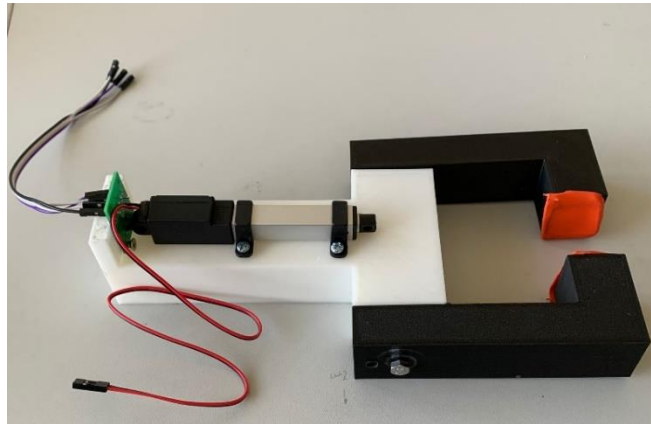


Figure 2. 3D printed mechanical clamp with embedded actuator and force sensor.



Figure 3. Mechanical coupler mounted on the test device.

Figure 4 shows the Unified Modelling Language (UML) Sequence Diagram, which represents the interaction amongst the different components of the system.

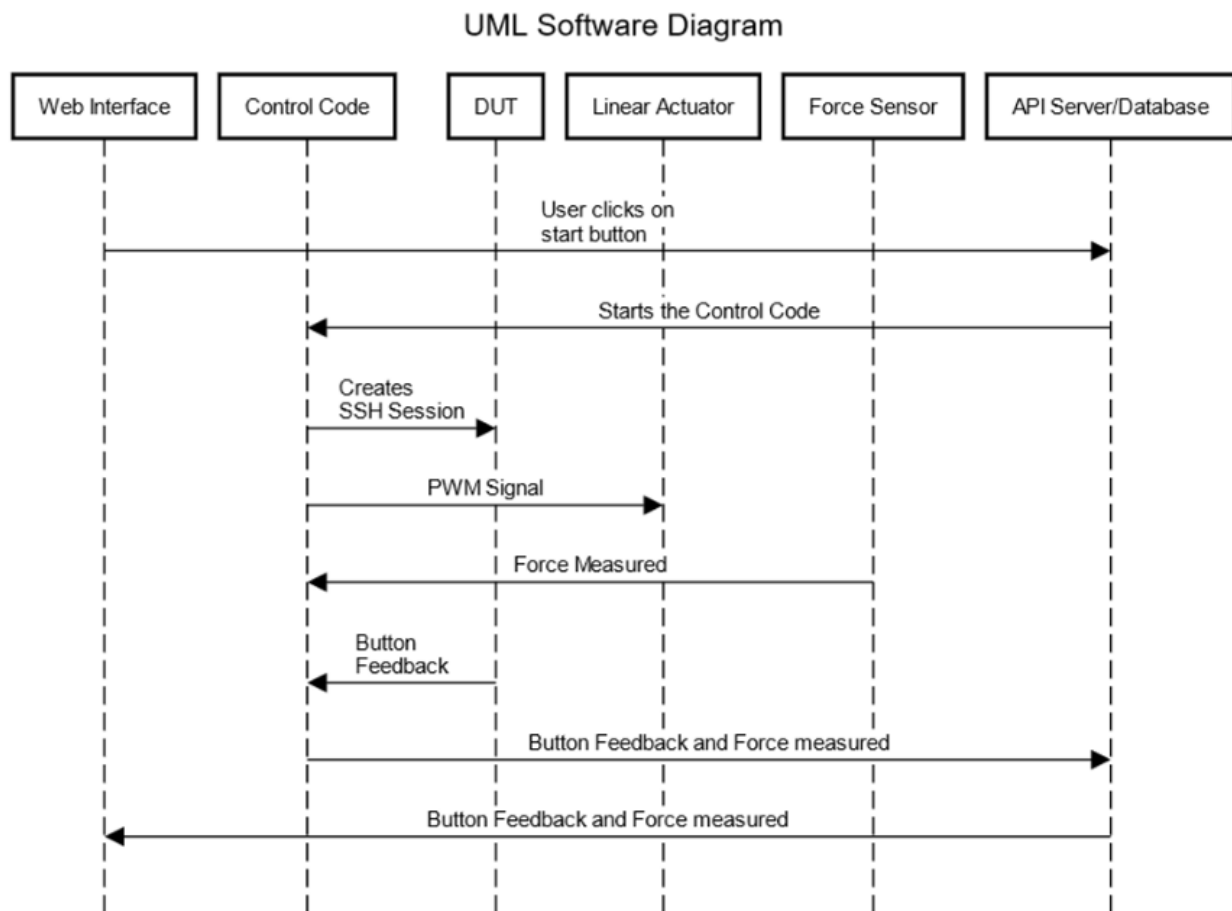


Figure 4. UML Diagram.

Further details on the system’s architecture can be found in the attached document “BlockDiagrams”.

Key milestones

ID	Description	Original planned completion date	Current planned completion date	Actual completion date
Communication with Raspberry Pi	Generate test signals and process feedback signals from the DUT.	04/12/2024	--/--/--	27/11/2024
Design of the mechanical attachment	Development of the mechanical couple using 3D print.	04/12/2024	--/--/--	5/12/2024

Individual tests to the main components	Functionality tests for the main components of the project: force sensor and linear actuator.	04/12/2024	--/--/--	28/11/2024
Integration of the actuator and sensor into the mechanical design	Mount the linear actuator with force sensor placed immediately behind it on the 3D-printed mechanical clamp	18/12/2024	18/12/2024	18/12/2024
Testing basic actuator control using the Raspberry Pi	Test the actuator's response and excursion against the code built in Python.	18/12/2024	18/12/2024	18/12/2024

Progress and deviation from plan

The only hitch we encountered after soldering the printed circuit board was that the team had soldered a connector with a diameter of 2.5mm for the 12V, 3A voltage source, which was incompatible with it. To make up for this, it was simply necessary to unsolder the wrong connector and solder the correct connector with a diameter of 2.1mm.

Work plan for next iteration

In the next iteration of the project, significant improvements will be made in various areas to improve the system's performance and functionality. In the Web Interface, the focus will be on organizing graphics and information, ensuring a clear presentation of statistics and test history, making it easier for users to analyse the data. In the mechanical coupling, we will work to integrate all the physical parts in a more cohesive way, resulting in a more elegant and discreet structure, suited to the aesthetic and functional requirements of the project. Regarding the DUT's feedback information, a more detailed system will be implemented that includes data such as the number of button presses during tests, offering greater visibility into the behaviour of the device in use. The current limiter will also be optimized with improvements in the sensitivity and robustness of the circuit, ensuring reliable operation even in demanding conditions. These advances aim to strengthen the overall quality of the project, making it more efficient and user-friendly. Finally, the auxiliary accessory for the reset buttons will also be developed.

Figure 5 shows the latest version of the 3D piece that the team intends to develop. In order to balance the force and inertia exerted by the actuator when pressing a button, which caused the weight to tip downwards and become misaligned, it was thought to immerse the actuator in the mechanical part and not leave it on its surface, so that the force is exerted at the centre point of the overall part. Figure 6 shows the updated sprint plan.

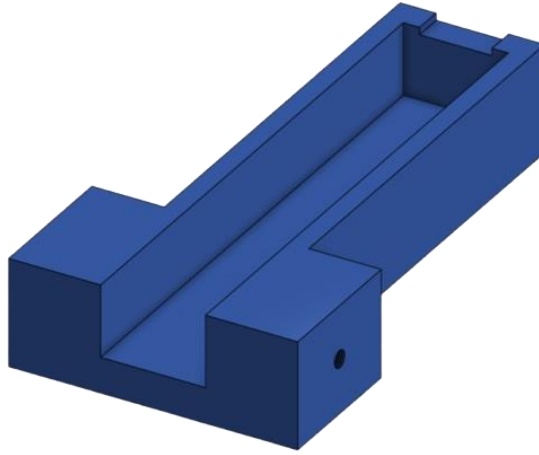


Figure 5. Latest version of the 3D part to be implemented in the next interaction.

Sprints	Weeks	Tasks		Claúdia	Rafael	Pedro	Gonçalo
1	4	Detailed system architecture and list of components finalized		x	x	x	x
		PCB design for the control signal and the Force Sensor	Web page development and structurization of a control code on the Rpi	x	x	x	x
MVP's		Web page interface and all PCB's					
2	3	Prototype a 3D support	Test basic Actuator control and SSH connection to the gateway with the RPi	x	x	x	x
			Test the controller and sensor PCB's	x	x	x	x
			Merge everthing into a single control code	x	x	x	x
MVP's		3D Support and Control Code		We are currently here			
3	2	Develop a 3D model to support the architecture	Update the interface to match the control code	x	x	x	x
		Test the overall architecture		x	x	x	x
4	2	Collect final feedback before final construction		x	x	x	x
		Delivery of the prototype to the client		x	x	x	x

Figure 6. Updated sprint plan.

Financial status

With respect to the project budget and components list, no modifications have been made since the previous milestone.

Quantity	Description		Status	Price Per Unite(€)	Total Value(€)	%
1	Honeywell FMAMSDXX015WC2C3	Force Sensor	Purchased	33,67	33,67	18%
1	Actuonix L12-50-50-12-S	Actuator	Purchased	83,66	83,66	45%
2	JST Comercial RE-H022TD-1190	Actuator connector to PCB	Purchased	0,18	0,36	0%
1	CUI Inc. SMM30-12-K-P6	Power Supplier	Purchased	17,76	17,76	10%
1	Same Sky PJ-079BH	Connector to Power Supplier	Purchased	1,5	1,5	1%
1	Raspberry Pi 4 B	Processing Unit	Purchased	32,55	32,55	17%
1	ST L6205D	Controler for the actuator	Purchased	6,12	6,12	3%
1	Nichicon GYF1H101MCQ1GS	Capacitor	Purchased	1,11	1,11	1%
1	KYOCERA AVX KAM21BR71H104JT	Capacitor	Purchased	0,19	0,19	0%
1	TAIYO YUDEN MAASU21GSB7224KTCA01	Bootstrap Capacitor	Purchased	0,4	0,4	0%
1	KYOCERA AVX KGF15AR71H103KT	Bootstrap Capacitor	Purchased	0,23	0,23	0%
2	KYOCERA AVX 12065A562FAT2A	Capacitor	Purchased	0,94	1,88	1%
2	onsemi / Fairchild 1N4148	Bootstrap Diode	Purchased	0,09	0,18	0%
2	Panasonic ERA-8VRB1003V	Resistor	Purchased	1,18	2,36	1%
1	Vishay CRCW2010100RJNEFIF	Bootstrap Resistor	Purchased	0,55	0,55	0%
1	TI INA381A3	Current Amplifier	Purchased	1,1	1,1	1%
1	TE Connectivity RLC73K2ER162FTDF	Current Measuring Resistor	Purchased	0,46	0,46	0%
1	Vishay MCS0402MD1002BE100	Pull-Up Resistor	Purchased	0,45	0,45	0%
1	CTS Electronic Components 14TR08SA103CP	Potentiometer	Purchased	0,71	0,71	0%
1	TI TLV9032QDGKRQ1	Capacitor	Purchased	1,5	1,5	1%
1	KYOCERA AVX KAM21BR71H104JT	Capacitor	Purchased	0,19	0,19	0%
				Total:	186,93	100%

Figure 7. Updated project budget and final components table.

Group contribution

Group member	Major contributions	Work share (%)
Cláudia Almeida	Team coordinator, responsible for organizing regrouping sessions and managing communication with external parties. Primary editor of written documents, developed the use cases, studied market relevance for industrial automation. Led Milestone 1 and Milestone 3 presentation. Web interface Development. 3D design and printing. Testing basic actuator control using the Raspberry Pi.	25
Rafael Alves	Developed the final hardware components list, analysed project risks for each phase and proposed mitigation strategies. Led Milestone 1 and Milestone 3 presentation. Development of the PCB. 3D design and printing. Testing basic actuator control using the Raspberry Pi.	25
Gonçalo Peralta	Created all system diagrams and drafts of the mechanical product for documentation, developed the block diagrams and use cases, sprints spread calendar and led the initial planning for the software approach to the project. Web interface Development. Led Milestone presentation 3. Communication with Raspberry Pi. Individual	25

	tests to the main components and testing basic actuator control using the Raspberry Pi.	
Pedro Matias	Contributed to researching and finalizing the hardware components list and supported the assessment of technical needs for project phases. Development of the PCB. Individual tests to the main components and testing basic actuator control using the Raspberry Pi.	25

Lessons Learned

Throughout the development of this project, several valuable lessons were learned, contributing to the improvement of processes and the success of the team. One of the main lessons was the importance of **parallelism between tasks**, which allowed different aspects of the project to be worked on simultaneously, speeding up progress without compromising quality. To support this approach, we adopted a **dynamic division of teams**, which proved essential for quickly adapting human resources to the needs of each project phase, guaranteeing efficiency and flexibility.

The implementation of **short daily stand-up meetings**, lasting between 5 and 10 minutes, was also a crucial element. These meetings fostered communication between team members, ensuring continuous alignment and rapid problem identification and resolution. Another lesson learned was **adaptability** in the face of unforeseen events. As unexpected challenges arose, it became clear that the ability to recalibrate plans and priorities in real time and to not go into a state of panic is indispensable for the success of complex projects.

In addition, we learned **the importance of a good understanding** of the project's objective right from the start, with regular alignments between the team and the advisors. These moments of harmony ensured that everyone involved had a clear and unified vision, reducing ambiguity and rework. Finally, the **importance of detailed and coherent documentation** became clear. It served as a reliable reference guide, facilitating both the monitoring of progress and the continuation of work at later stages. These lessons, as well as optimizing the execution of the project, will be fundamental pillars in future projects.